

Ashrut Aryal

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EDUCATION

Michigan State University

PhD, Mechanical Engineering, GPA: 3.88/4.0

MS, Mechanical Engineering

Kathmandu University

BS, Mechanical Engineering

East Lansing, MI

Expected May 2027

Expected May 2026

Kathmandu, Nepal

2017 - 2021

RESEARCH EXPERIENCE

Graduate Research Assistant

Michigan State University, MOTRE Lab

Jan 2024 - Present

Autonomous Robotic System for Cloth Flattening

- Developed system where robot learns to flatten deformed cloth by observing human demonstrations; built perception system to detect wrinkles and extract geometric features (location, orientation, size)
- Implemented 3D pose estimation system using ArUco markers in Python to track finger motion during human subject study; used camera calibration to transform marker poses to world coordinates
- Trained behavior cloning models (neural network, linear regression) to map wrinkle geometry to manipulation actions; discovered linear models generalized better than neural networks with limited data (~100 demonstrations)
- Integrated real-time perception with robot control on FANUC industrial manipulator to autonomously flatten cloth using continuous visual feedback

Body-Machine Interface for Industrial Sewing Machine Control

- Developed a torso-lean IMU Body-Machine Interface (BoMI), enabling precise control of industrial sewing machine (continuous speed control + discrete state control)
- Interfaced sewing machine control board via Arduino (UNO R4 DAC) to output pedal-equivalent analog voltages, replacing foot pedal with IMU-based torso lean control, incorporated haptic feedback

WORK EXPERIENCE

Robotics and AI Engineer

National Innovation Center, Kathmandu, Nepal

June 2022 - July 2023

- Developed real-time multi-object tracker using YOLO detection with SORT architecture, Kalman filtering for state estimation and track continuity during occlusions, and Hungarian algorithm for IOU-based data association
- Built synchronized multi-sensor data acquisition system (dual RGB-D, LiDAR, odometry) with PyQt annotation GUI

PROJECTS

Behavior Cloning for Robotics Reaching

- Built learning pipeline in PyBullet: Cartesian teleoperation with inverse kinematics, MLP policy via behavior cloning, DAGger for distribution shift; 85% success on novel targets. Experience with MuJoCo simulation

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C, C++, Arduino

Computer Vision & ML: OpenCV, YOLO, PyTorch, TensorFlow, ArUco markers, 3D pose estimation, Kalman Filter

Robotics & Simulation: ROS 2, FANUC Robot, PyBullet, MuJoCo, SolidWorks, PyQt

PUBLICATIONS

- A. Aryal, N. Kant, R. Ranganathan, R. Mukherjee, C. Owen, "Robotic Manipulation for Flattening Wrinkled Cloth: A Human-Inspired Algorithm," *IEEE Transactions on Human-Machine Systems*, 2025 (Under Review)
- N. Kant, A. Aryal, R. Ranganathan, R. Mukherjee, C. Owen, "Modeling Human Strategy for Flattening Wrinkled Cloth Using Neural Networks," *IEEE SMC*, 2024. <https://doi.org/10.1109/SMC54092.2024.10832048>
- R. Ranganathan, A. Aryal et al., "Control Strategies for High-DOF Assistive Devices Using Non-Invasive Body-Machine Interfaces," *Disabil. Rehabil.: Assist. Technol.*, 2025. (Accepted)